

SAI HARSHA VEMULA

Tuscaloosa City, Alabama 35401

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DevOps Engineer specializing in infrastructure automation, container orchestration, and distributed system reliability. Experienced in developing backend services with Java and Python, provisioning cloud environments with Terraform and Ansible, and delivering production-grade CI/CD workflows across AWS and Kubernetes ecosystems.

Education

University of Missouri-Kansas City

Master in Computer Science

August 2023 – December 2024

Kansas City, Missouri

Vellore Institute of Technology

M Tech Integrated Software Engineering

June 2016 – June 2021

Vellore, Tamil Nadu, India

Technical Skills

Languages: Python, Shell Scripting, SQL, CSS, JavaScript

Database Technologies: MySQL, PostgreSQL, Elasticsearch, Redis

DevOps Tools & CI/CD: GitLab, Jenkins, SonarQube, Argo CD, Velero

Containerization and Orchestration: Docker, Kubernetes

AWS Cloud Services: EKS, EC2, IAM, RDS, Lambda, VPC, Cloudwatch, SNS, ECS, Step Functions, WAF

Azure Cloud Services: Azure Virtual Machines, Azure App Service, AKS, Azure Functions, Azure AD

Security & Secrets Management: HashiCorp Vault, Trivy, Grype, GitLeaks

Protocols: TCP, UDP, IP, HTTP, HTTPS, SSH, DNS, SFTP

Monitoring & Logging: Prometheus, Grafana, Datadog, ELK Stack (Elasticsearch, Logstash, Kibana)

Infrastructure as Code (IaC): Terraform, AWS CloudFormation, Ansible

Build Tools: Maven, Gradle

Web Servers: Tomcat, Nginx, Apache HTTP Server

Operating Systems: Linux (Ubuntu, CentOS), Windows

Experience

University of Alabama – Alabama Water Institute

DevOps Engineer

July 2023 – Present

Tuscaloosa, Alabama, USA

- Automated infrastructure creation for research data workflows using Terraform and AWS services including EC2, S3, Lambda, and Step Functions, improving reproducibility and reducing deployment time by **45%**.
- Implemented AWS security enhancements through CloudTrail, Security Hub, GuardDuty, and IAM Access Analyzer to ensure compliance with institutional and AWS security standards.
- Established automated vulnerability scanning using **Trivy**, **Grype**, and **SonarQube**, integrating scans into CI/CD workflows for continuous security validation.
- Developed automated build pipelines for multi-architecture Docker images (x86 and ARM64) using GitHub Actions and AWS CodeBuild, improving build consistency and testing coverage.
- Created detailed AWS architecture diagrams and documentation for system design, orchestration workflows, and infrastructure dependencies, improving team collaboration and knowledge sharing.
- Implemented automated backup and retention policies for S3, EBS, and RDS resources, ensuring data resilience and compliance with university IT governance standards.
- Optimized AWS resource utilization by auditing EBS volumes, EC2 instances, and S3 storage classes, achieving **30%** monthly cost savings.
- Enhanced observability with CloudWatch metrics and dashboards, integrating alerting pipelines for workflow executions and AWS health events.
- Collaborated with research and development teams to streamline model execution workflows across AWS environments using container orchestration and CI/CD automation.
- Contributed to automation testing pipelines for validating cloud infrastructure, ensuring environment reliability and reproducibility for hydrologic modeling frameworks.
- Integrated AWS Config and Trusted Advisor checks into continuous compliance workflows, ensuring real-time detection and remediation of policy violations across multi-account environments.
- Collaborated with cross-functional teams to design and deploy scalable CI/CD pipelines for scientific modeling workloads, enabling reproducible, event-driven research automation across cloud and HPC infrastructures.

Vitrana, Inc.

July 2021 – June 2023

Senior Software Engineer - DevOps

Bangalore, Karnataka, India

- Developed and maintained Kubernetes manifests for microservices, automating deployment and scaling workflows that improved deployment speed by **20%**.
- Applied Infrastructure as Code (IaC) using Terraform and Ansible to automate multi-environment provisioning, eliminating configuration drift and reducing setup time by **50%**.
- Established CI/CD pipelines integrating SonarQube, Trivy, and JUnit, enhancing code quality and reducing security vulnerabilities.
- Managed AWS networking and compute resources (VPC, subnets, EC2, RDS, WAF, IAM), strengthening security and reducing downtime by **20%**.
- Implemented centralized observability using Prometheus and Grafana, reducing incident response times by **40%**.
- Containerized legacy applications with Docker and improved release management using GitLab and Argo CD, streamlining deployments across environments.
- Optimized database performance and caching using Redis and AWS ElastiCache, enhancing application responsiveness and scalability.
- Created automated disaster recovery and backup workflows, improving data reliability and business continuity.
- Developed Python automation scripts to monitor CI/CD pipeline health and auto-trigger rebuilds on failure, reducing manual restarts and improving overall pipeline reliability by **35%**.

Vitrana, Inc.

August 2020 – July 2021

Associate Software Engineer

Bangalore, Karnataka, India

- Automated infrastructure and monitoring workflows using Bash, Python, and AWS Lambda, reducing manual intervention by **40%**.
- Led migration of on-prem applications to AWS, improving scalability and reducing operational costs by **15%**.
- Implemented IAM roles, RBAC policies, and Kubernetes access controls to enforce least privilege and ensure compliance.
- Deployed and maintained the ELK stack for centralized logging and analytics, accelerating issue resolution by **30%**.
- Developed Helm charts and automated Ansible playbooks for microservices deployment, improving consistency and release efficiency.
- Built Python utilities to automatically rotate and validate AWS IAM access keys across multiple environments, enhancing security posture and eliminating credential expiration-related outages.
- Built an analytics dashboard using AWS QuickSight and Redshift, providing real-time visibility into operational metrics.

Projects

Serverless Web Application | AWS Lambda, API Gateway, DynamoDB, S3

- Engineered a highly scalable serverless web application leveraging AWS Lambda and DynamoDB for seamless and low-latency CRUD operations, optimizing response times via AWS API Gateway.
- Integrated S3 for static website hosting, ensuring smooth frontend-backend interactions and an enhanced user experience with minimal operational overhead.

Microservices-Based Healthcare System | Java, Spring Boot, Docker, Kubernetes, AWS

- Developed a resilient and scalable microservices-based healthcare system using Java and Spring Boot, deploying containerized services via Docker and Kubernetes for enhanced scalability and load balancing.
- Optimized CI/CD pipelines with Jenkins and GitLab, automating deployments on AWS and cutting release time by **50%**, while implementing centralized logging with the ELK stack for improved observability and issue resolution.

Enterprise Cloud Migration and Optimization | AWS, Terraform, Kubernetes, GitLab CI/CD, Ansible

- Migrated enterprise applications from on-premise infrastructure to AWS, improving scalability and cutting operational costs by **40%**.
- Automated infrastructure provisioning and deployment using Terraform, Kubernetes, and GitLab CI/CD, reducing deployment time by **50%** and ensuring high availability.

Achievements & Certifications

AWS Certified Solutions Architect – Associate

Verified by Credly, Badge Link: [AWS Certified Solutions Architect – Associate](#)

HashiCorp Certified: Terraform Associate

Verified by Credly, Badge Link: [HashiCorp Terraform Associate](#)

Certified Kubernetes Administrator (CKA) with Practice Tests By Kodekloud

Cert Link: [Certified Kubernetes Administrator \(CKA\) with Practice Tests](#)

GitHub Actions

Cert Link: [GitHub Actions](#)

Microsoft Certified: Azure Fundamentals

Cert Link: [Microsoft Certified: Azure Fundamentals](#)